Assignment 5.1 Exercise

SHAP Loan Approval Explainability Analysis

In this exercise, you will analyze the loan approval dataset and utilize SHAP to understand feature contributions to model predictions. The goal is to interpret how specific features affect decisions and reflect on the ethical implications of such models.

Instructions:

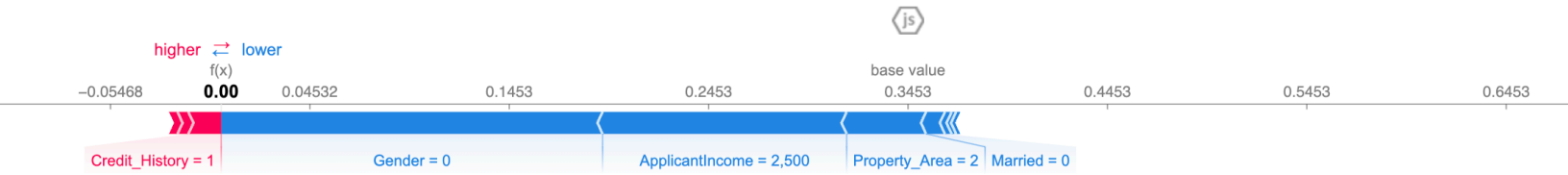
1. Complete “Step 3: SHAP Explanations” in the **assignment_5.1_notebook** copy and paste your output and respond to the questions into this exercise document.
2. Provide short responses to the reflection questions in “Step 4: Ethical Reflection”

Step 3: SHAP Explanations

Use SHAP to explain the predictions of your logistic regression model.

1a. Generate the SHAP force plot for the first data point in your test set using the SHAP values. The force plot should explain how individual features contribute to pushing the model’s prediction towards or away from class 1.

Paste the force plot here:



1b. Which features are pushing the prediction towards class 1? Which features are pushing the prediction away from class 1?

Response:

The features that are pushing the prediction towards class 1 are credit_history. The features that are pushing the prediction away from class 1 are gender, applicantIncome, property_area, and married.

2a. Create a bar plot that displays the SHAP values (feature contributions) for the second data point and SHAP values for class 1. The bar plot will show how much each feature contributes to the model's prediction for class 1, with the features ranked by their contribution.)

Paste the bar plot here:

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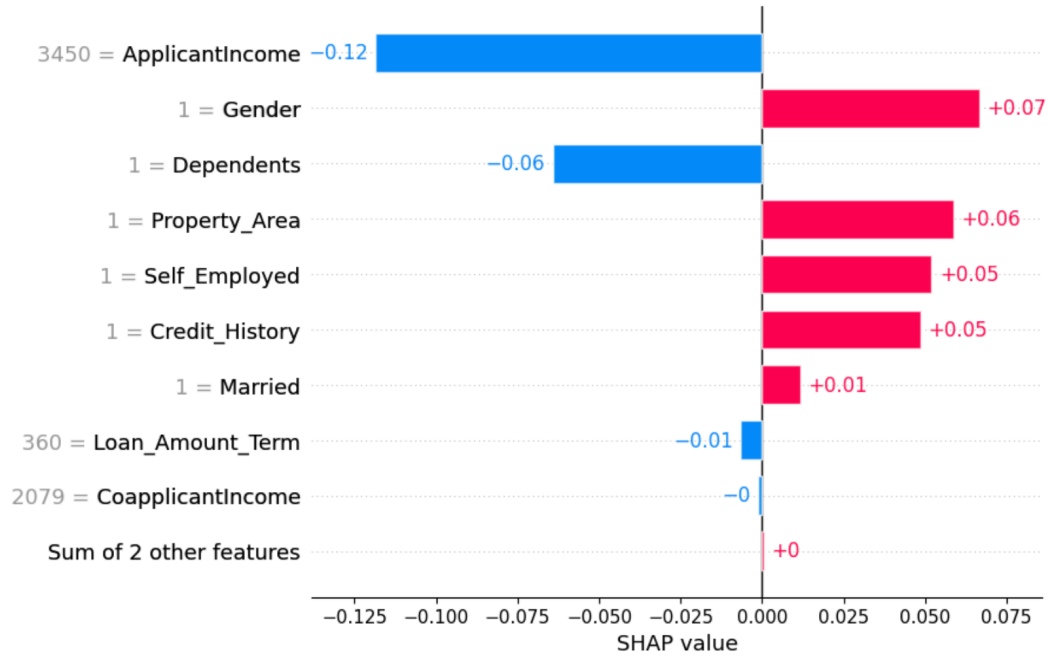

2b. Which feature had the greatest influence on the prediction for this data point? Were most features pushing the prediction toward or away from class 1? Are there any surprising variables that are influencing the prediction?

Response:

The feature with the greatest influence on the prediction for this data point was the ApplicantIncome, contributing a strong negative effect of -0.12 and pushing the prediction away from class . Most features exerted a positive influence, with variables such as Gender, Property_Area, Self_Employed, Credit_History, and Married nudging the prediction toward approval, indicating that the model was generally inclined toward class 1. However, this positive trend was partially offset by a few negatively contributing features, particularly ApplicantIncome and Dependents and Loan_Amount_Term. The negative impact of ApplicantIncome is somewhat counterintuitive, as higher income would typically support loan approval. Additionally, the positive contribution of Gender raises ethical concerns, as it implies that demographic attributes may be influencing decisions, highlighting potential bias and the need for fairness evaluation in the model.

3a. Create a beeswarm plot to visualize the global importance of features.

Paste the beeswarm plot here:

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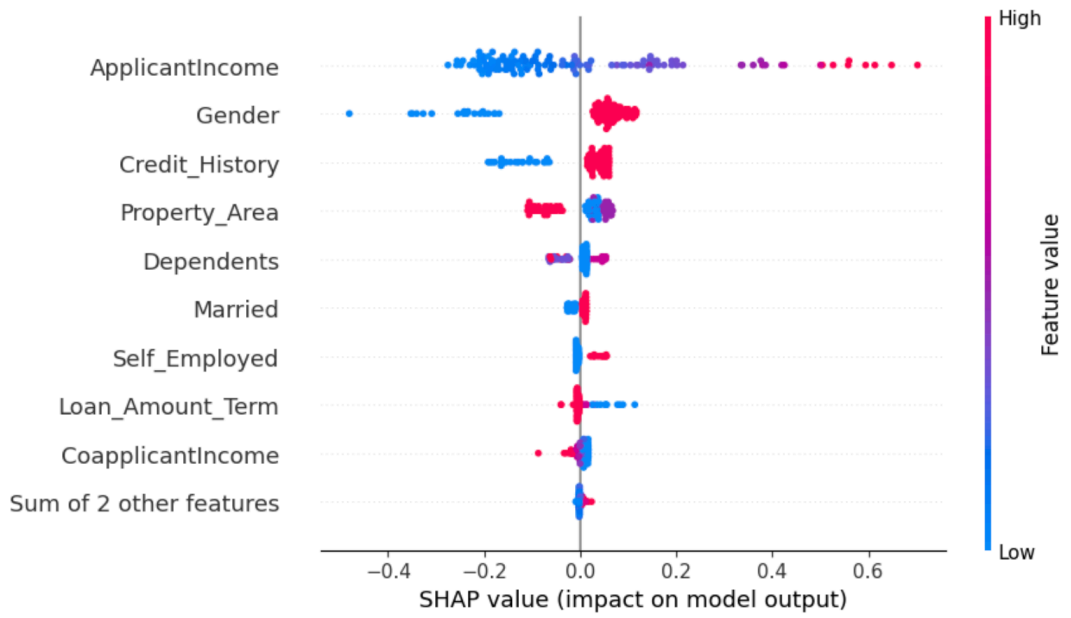

3b. Which feature appears to be the most important across all data points? Are there any features that have a mixed effect (i.e. pushing predictions towards class 1 and other times pushing them away?) What could cause this variability? How do the color of the dots help you interpret the relationship between the feature value and the prediction? Are there any notable findings?

Response:

The feature that appears to be the most important across all data points is the ApplicationIncome, since it has the widest spread and is ranked at the top. Several features exhibit a mixed effect, including ApplicantIncome, Property_Area, Dependents, and CoapplicantIncome, where their SHAP values span both positive and negative ranges. This variability is typically caused by feature interactions, nonlinear relationships, or differing contexts within the data; for example income may help approval at moderate levels but hurt at very high or low levels depending on other factors.

The color of the dots represents the feature value (blue = low, red = high), which helps interpret directionality. For example, in ApplicantIncome, higher values (red) tend to push predictions toward approval (positive SHAP values), while lower values (blue) push them away, indicating a generally positive relationship. Similarly, for Credit_History, higher values strongly push toward approval, while lower values push away, showing a clear and consistent pattern.

Notable findings include the strong and consistent positive impact of Credit_History, as well as the presence of mixed effects in several features, suggesting the model captures complex relationships rather than simple linear rules. Additionally, features like Gender still show some influence, which may raise fairness and bias concerns, as demographic attributes should ideally not significantly affect loan decisions.

Step 4: Ethical Reflection

As you explored the SHAP plots, you were able to see how individual features contribute to a model's predictions, increasing transparency. However, even with these interpretability tools, there are broader ethical concerns regarding bias and transparency in machine learning systems.

Machine learning models, particularly those used in high-stakes areas like healthcare, finance, and criminal justice, can inadvertently reinforce societal biases present in the training data (e.g., based on race, gender, income). While tools like SHAP provide insights into how models make decisions, they do not prevent biased outcomes.

1. How can explainability tools such as SHAP help identify and address bias in machine learning models?

Your response:

Explainability tools like SHAP help identify bias in machine learning models by revealing how individual features influence predictions. By quantifying each feature's contribution, SHAP can highlight when sensitive attributes (such as gender or proxies like location or income) disproportionately affect outcomes. This allows practitioners to detect both direct and indirect bias, as well as compare how features behave across different demographic groups to uncover unfair patterns.

Although SHAP does not remove bias, it provides the transparency needed to address it. Insights from SHAP analyses can guide corrective actions such as removing problematic features, rebalancing training data, or applying fairness constraints. In this way, explainability tools support the development of more accountable and equitable machine learning systems.

2. How does explainability contribute to transparency in machine learning systems?

Your response:

Explainability enhances transparency in machine learning systems by making otherwise "black-box" models more understandable to humans. It reveals how input features influence predictions, allowing users to see why a model made a particular decision rather than just the outcome. This helps stakeholders, such as developers, regulators, and end users, gain insight into the model's logic, assumptions, and decision patterns.

By providing clear explanations of feature contributions and model behavior, explainability also supports accountability and trust. It enables practitioners to validate whether the model is using appropriate factors, detect errors or biases, and justify decisions in high-stakes applications like finance or healthcare. Thus, explainability bridges the gap between complex algorithms and human understanding, making machine learning systems more transparent, interpretable, and responsible.